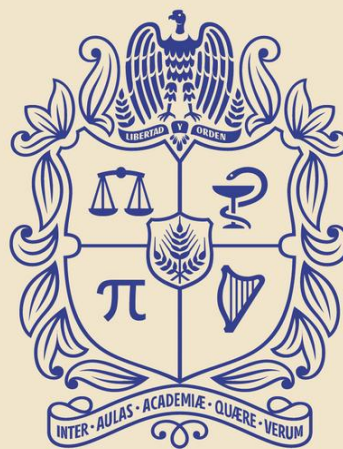




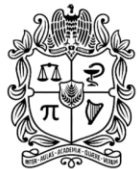
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Numerical Differentiation and Integration

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1. Introduction



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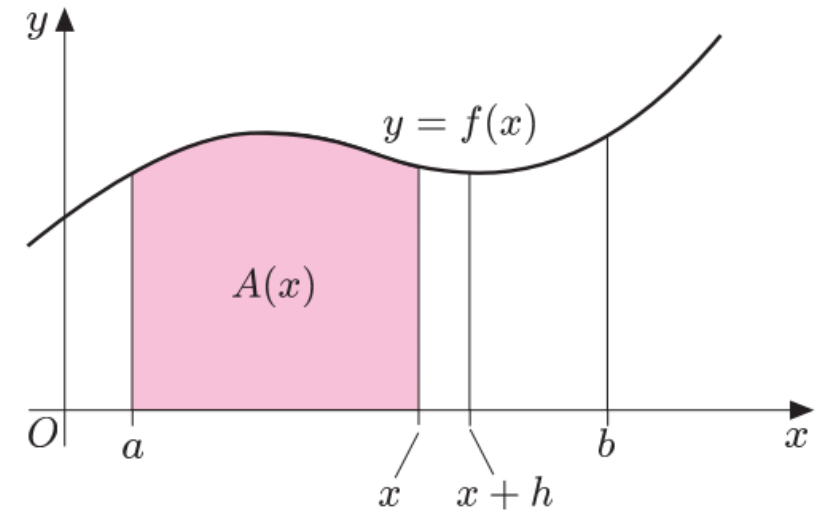
Overview

Differentiation

- The first derivative gives the rate of change (dy/dx).
- The second derivative measures curvature.
- Partial derivatives describe slopes on multivariable surfaces.

Integration

- The inverse of differentiation: $\int_a^b f(x)$ represents area under the curve.
- Evaluating an integral is equivalent to solving $dy/dx = f(x)$ with an initial condition.





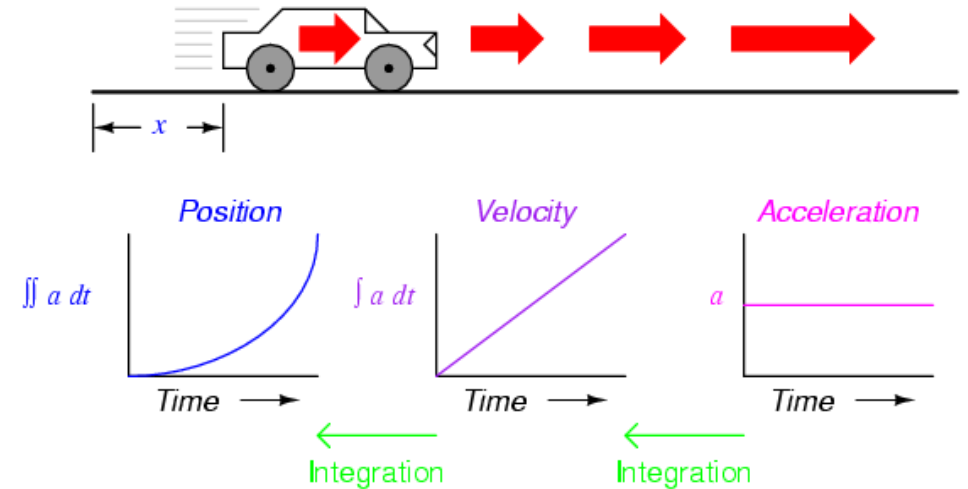
Overview

Derivative–Integral Duality

- Differentiate position $\rightarrow$ velocity; integrate velocity $\rightarrow$ position.
- Both processes are complementary and framed as differential equations.

Function Types in Numerical Work

- Simple continuous (e.g., polynomials).
- Complex continuous (hard to treat analytically).
- Tabulated data (common in experiments).

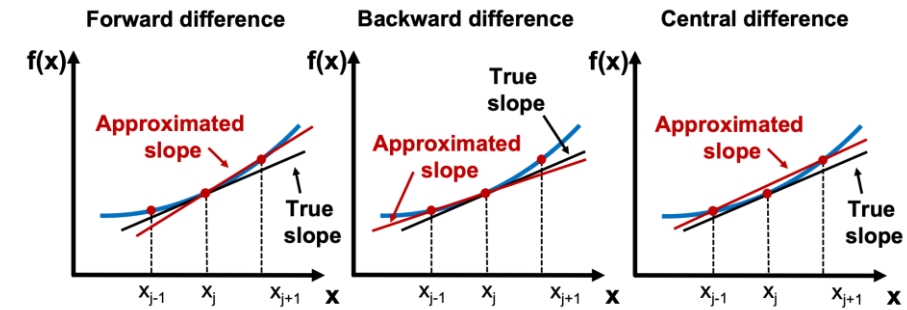




Overview

Key Numerical Techniques

- Finite differences for derivatives.
 - Quadrature formulas (refined strip methods) for integrals.
 - Curve-fitting plus differentiation for noisy data.
-
- **Engineering Applications**
 - Newton's law ($F = ma$), Fourier's law ($q = -k' dT / dx$), fluid dynamics, mass/energy transfer.
 - Integrals for continuous means, total mass, centers of gravity, RMS current, total heat, and more.





2. Newton-Cotes Integration Formulas

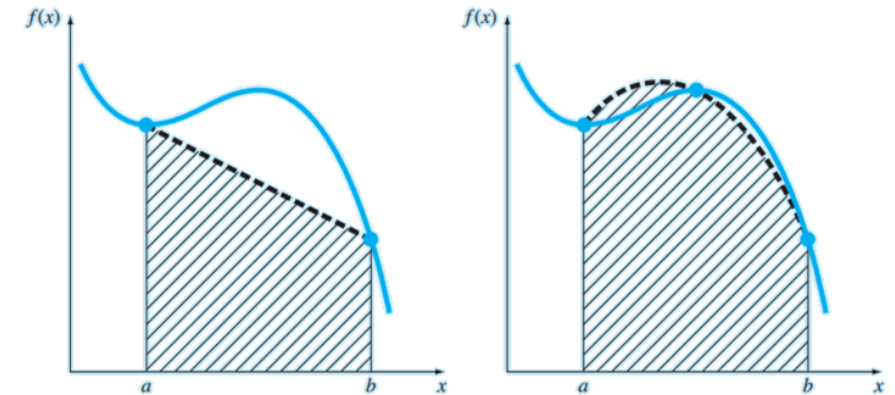


Newton–Cotes strategy

Replace $f(x)$ over $[a,b]$ by a polynomial $f_n(x) = a_0 + a_1x + \cdots + a_nx^n$, then integrate the polynomial instead of the original function.

Closed vs. open forms

- *Closed*: use data at both endpoints a and b .
- *Open*: integration limits extend beyond the data range (rarely used for definite integrals).





Trapezoidal Rule

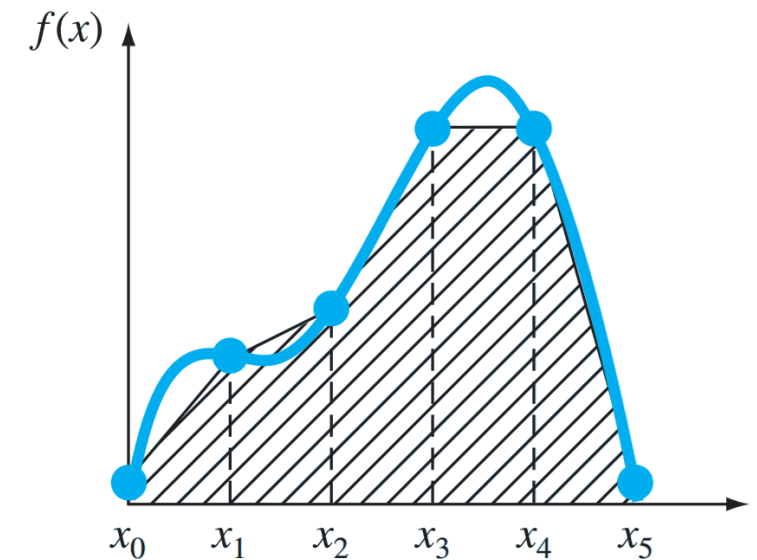
Single-segment Trapezoidal Rule

$$I = \int_a^b f(x)dx \approx (b - a) \frac{f(a) + f(b)}{2}$$

Composite (multiple-application) Trapezoidal Rule

- Divide $[a, b]$ into n equal subintervals of width $h = (b - a)/n$.
- Approximate each with the trapezoidal rule and sum:

$$I \approx \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right]$$





Example 1

Use the two-segment trapezoidal rule to estimate the integral of

$$f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$$

from $a = 0$ to $b = 0.8$. Recall that the correct value for the integral is 1.640533.



Simpson's rules

Idea of Simpson's rules

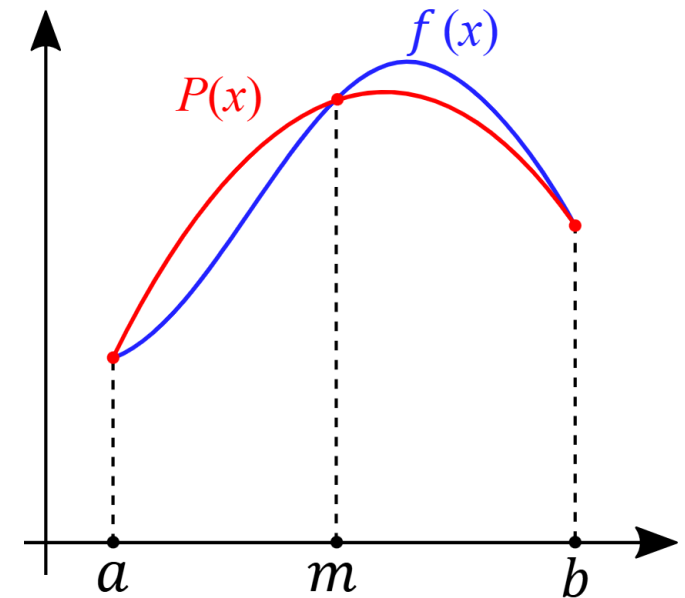
Use higher-order interpolating polynomials (parabolas, cubics) through equally spaced points to approximate $\int_a^b f(x)dx$.

Simpson's 1/3 Rule (single segment)

- Divide $[a, b]$ at midpoint $x_1 = (a + b)/2$, with $h = (b - a)/2$.
- Formula:

$$\begin{aligned} I &\approx \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] \Leftrightarrow I \\ &\approx \frac{(b-a)}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right] \end{aligned}$$

- Exact for any cubic f .





Simpson's rules

Composite Simpson's 1/3 Rule

- Split $[a,b]$ into n even-numbered subintervals of width $h=(b-a)/n$.
- Apply 1/3 rule on each adjacent pair and sum:

$$I \approx \frac{h}{3} \left[f(x_0) + 4 \sum_{i \text{ odd}} f(x_i) + 2 \sum_{i \text{ even}, i \neq 0, n} f(x_i) + f(x_n) \right]$$



Example 2

Use Composite Simpson's 1/3 Rule with $n = 4$ to estimate the integral of

$$f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$$

from $a = 0$ to $b = 0.8$. Recall that the correct value for the integral is 1.640533.



Simpson's rules

Simpson's 3/8 Rule (single segment)

Use four points $x_0, a, x_1 = a + h, x_2 = a + 2h, x_3 = b$, with $h = (b - a)/3$.

Formula:

$$I \approx \frac{3h}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)] \Leftrightarrow I \approx \frac{b-a}{8} \left[f(a) + 3f\left(a + \frac{b-a}{3}\right) + 3f\left(a + \frac{2(b-a)}{3}\right) + f(b) \right]$$

Slightly more accurate than single 1/3 rule but requires an odd number of segments.

Combining 1/3 and 3/8

For an overall odd number of segments: apply 1/3 rule on all but the last three, and 3/8 rule on the final three to maintain third-order accuracy.



Example 3

Use Simpson's 3/8 rule to integrate

$$f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$$

from $a = 0$ to $b = 0.8$.

Use it in conjunction with Simpson's 1/3 rule to integrate the same function for five segments.



3. Numerical Differentiation



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High-accuracy Differentiation Formulas

What are these formulas for?

When you know the value of a function $f(x)$ only at equally spaced points $x_i, x_{i\pm 1}, x_{i\pm 2}, \dots$ (spacing h), you can **approximate its derivatives** by replacing Taylor-series terms with finite-difference quotients.

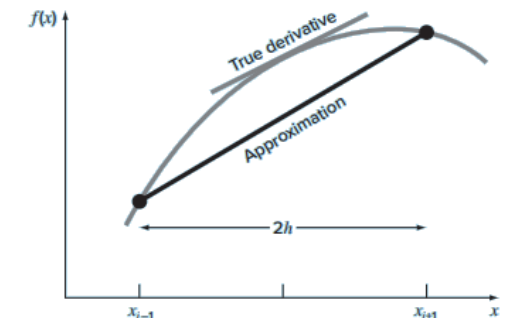
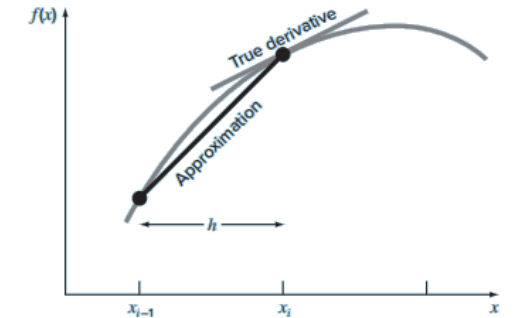
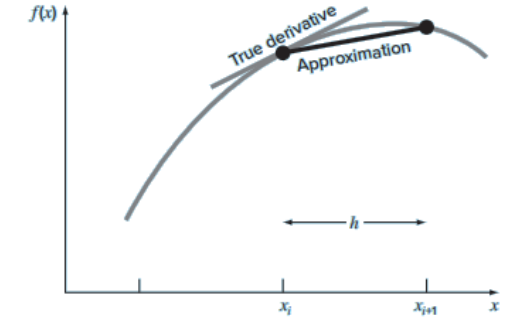
- The “**short**” **formulas** use the minimum number of points and keep only the leading error term.
- The “**long**” or “**high-accuracy**” **formulas** include additional Taylor terms and combine several f -values so that lower-order errors cancel out—hence the higher accuracy.



High-accuracy Differentiation Formulas

Three families of schemes

Family	Where it samples f	Typical use
Forward	$x_i, x_{i+1}, x_{i+2}, \dots$	Left boundary of the grid when no points exist to the left.
Backward	$x_i, x_{i-1}, x_{i-2}, \dots$	Right boundary—mirror image of forward.
Centered	Symmetrically $x_{i\pm 1}, x_{i\pm 2}, \dots$	Interior nodes; generally the most accurate.





High-accuracy Differentiation Formulas

Key formulas (up to the fourth derivative)

Compact notation: $f_k = f(x_k)$

Derivative	Forward	Backward	Centered
1st	$\frac{f_{i+1} - f_i}{h} O(h)$ $\frac{-f_{i+2} + 4f_{i+1} - 3f_i}{2h} O(h^2)$	$\frac{f_i - f_{i-1}}{h} O(h)$ $\frac{3f_i - 4f_{i-1} + f_{i-2}}{2h} O(h^2)$	$\frac{f_{i+1} - f_{i-1}}{2h} O(h^2)$ $\frac{-f_{i+2} + 8f_{i+1} - 8f_{i-1} + f_{i-2}}{12h} O(h^4)$
2nd	$\frac{f_{i+2} - 2f_{i+1} + f_i}{h^2} O(h)$ $\frac{-f_{i+3} + 4f_{i+2} - 5f_{i+1} + 2f_i}{h^2} O(h^2)$	$\frac{f_i - 2f_{i-1} + f_{i-2}}{h^2} O(h)$ $\frac{2f_i - 5f_{i-1} + 4f_{i-2} - f_{i-3}}{h^2} O(h^2)$	$\frac{f_{i+1} - 2f_i + f_{i-1}}{h^2} O(h^2)$ $\frac{-f_{i+2} + 16f_{i+1} - 30f_i + 16f_{i-1} - f_{i-2}}{12h^2} O(h^4)$
3rd	$\frac{f_{i+3} - 3f_{i+2} + 3f_{i+1} - f_i}{h^3} O(h)$ $\frac{-3f_{i+4} + 14f_{i+3} - 24f_{i+2} + 18f_{i+1} - 5f_i}{2h^3} O(h^2)$	$\frac{f_i - 3f_{i-1} + 3f_{i-2} - f_{i-3}}{h^3} O(h)$ $\frac{5f_i - 18f_{i-1} + 24f_{i-2} - 14f_{i-3} + 3f_{i-4}}{2h^3} O(h^2)$	$\frac{f_{i+2} - 2f_{i+1} + 2f_{i-1} - f_{i-2}}{2h^3} O(h^2)$ $\frac{-f_{i+3} + 8f_{i+2} - 13f_{i+1} + 13f_{i-1} - 8f_{i-2} + f_{i-3}}{8h^3} O(h^4)$
4th	$\frac{f_{i+4} - 4f_{i+3} + 6f_{i+2} - 4f_{i+1} + f_i}{h^4} O(h)$ $\frac{-2f_{i+5} + 11f_{i+4} - 24f_{i+3} + 26f_{i+2} - 14f_{i+1} + 3f_i}{h^4} O(h^2)$	$\frac{f_i - 4f_{i-1} + 6f_{i-2} - 4f_{i-3} + f_{i-4}}{h^4} O(h)$ $\frac{3f_i - 14f_{i-1} + 26f_{i-2} - 24f_{i-3} + 11f_{i-4} - 2f_{i-5}}{h^4} O(h^2)$	$\frac{f_{i+2} - 4f_{i+1} + 6f_i - 4f_{i-1} + f_{i-2}}{h^4} O(h^2)$ $\frac{-f_{i+3} + 12f_{i+2} - 39f_{i+1} + 56f_i - 39f_{i-1} + 12f_{i-2} - f_{i-3}}{6h^4} O(h^4)$



Example 4

Estimated the derivative of at $x = 0.5$ using finite divided differences and a step size of $h = 0.25$

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$$



Derivatives of Unequally Spaced Data

Lagrange Interpolating Polynomial

Fit a second-order Lagrange polynomial to any three adjacent points (no equal spacing required) and differentiate it analytically.

Resulting Expression (Three-Point Formula)

$$f'(x) = f(x_{i-1}) \frac{x_i - x_{i+1}}{(x_{i-1} - x_i)(x_{i-1} - x_{i+1})} + f(x_i) \frac{2x_i - x_{i-1} - x_{i+1}}{(x_i - x_{i-1})(x_i - x_{i+1})} + f(x_{i+1}) \frac{x_i - x_{i-1}}{(x_{i+1} - x_{i-1})(x_{i+1} - x_i)}$$



Example 4

Derivatives of unequally spaced data via the second-order Lagrange interpolating polynomial (three-point formula).

A thermocouple records the temperature T (in $^{\circ}\text{C}$) of a reactor at unevenly spaced times.

The measured values are:

t (s)	T ($^{\circ}\text{C}$)
4.0	87.2
5.1	92.8
7.0	104.5



4. Summary



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Summary

- **When analytic work fails.** Numerical integration and differentiation are invoked when a function is too complicated—or available only as discrete data.
- **Newton–Cotes family.** Trapezoidal and Simpson ($1/3$ & $3/8$) rules approximate areas by fitting low-degree polynomials; higher degree $\Rightarrow$ higher accuracy.
- **Finite-difference formulas.** Forward, backward and centred stencils estimate derivatives; expanding the stencil order reduces truncation error.
- **Uneven spacing & trade-offs.** Lagrange interpolation handles irregular data, but engineers must balance accuracy, cost, and data resolution when choosing a scheme.